

Optimal Portfolio Construction & Risk-Return Analysis

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Executive Summary

In this report, the historical returns of Exxon Mobil Corporation (XOM), Johnson & Johnson (JNJ), Advanced Micro Devices, Inc. (AMD) are analysed and compared against the Market Index to give specific assets allocation recommendations. As per the given tasks, we computed individual expected returns, variances and Sharpe ratios, culminating in the construction of an optimal portfolio and comparing it with the Market Index to determine the best risk-return ratio.

Based on our analysis, AMD generated the highest return, with an arithmetic mean rate of return of 2.1033%. It is followed by the Market Index (0.7550%), XOM (0.5280%), and JNJ (0.4355%).

In terms of Sharpe ratio, the Market Index was the highest (0.1447), followed by AMD (0.1166), JNJ (0.0717) and XOM (0.0600). This means that the Market Index is the most efficient stock amongst the four in terms of risk to reward.

A portfolio comprising XOM, JNJ, and AMD was constructed and benchmarked against the market index. The optimal portfolio weights for the three stocks were determined through the calculation of expected returns, variances, and covariances. Subsequently, the Capital Allocation Line (CAL) was used to form a combined portfolio of risky and risk-free assets, generating higher expected returns while maintaining the same level of risk (standard deviation).

Using the regression model, the portfolio's alpha was found to be slightly positive (0.0015) while its beta was marginally above 1 (1.0998). This indicates that the portfolio is more sensitive to market movements but is also capable of delivering returns that exceed those predicted by its level of risk. In this case, the positive alpha highlights its superior risk-adjusted performance relative to the market. Hence, this portfolio would appeal to investors seeking higher potential returns and are willing to tolerate greater exposure to market risk.

TASK 1: Graph the stock price levels

Historical Performance of Companies Share Price (31/12/04 – 31/12/24)

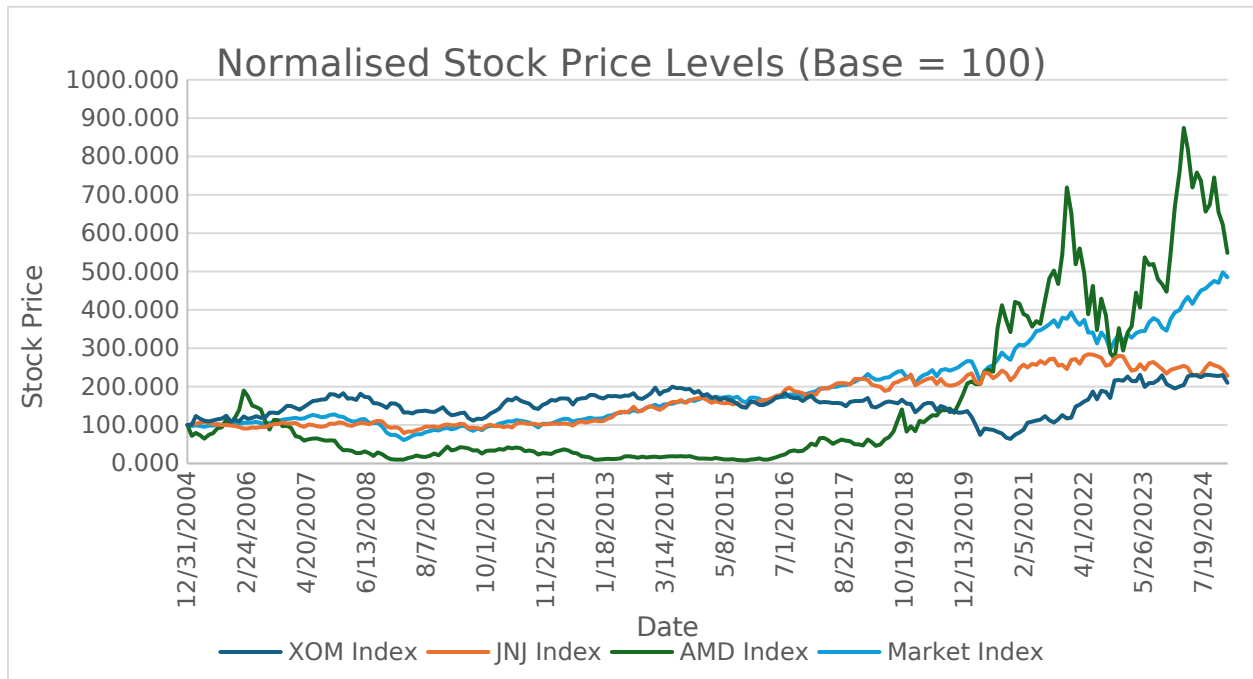


Fig 1.1: Normalised stock price levels for AMD, JNJ, XOM and Market Index

Periods of time with unusually large price changes



Fig 1.2: Normalised Market Index Return

Year	Event	Description
2007 - 2009	2008 Global Financial Crisis	The 2008 American mortgage crisis triggered a worldwide financial collapse, with Lehman Brother’s bankruptcy on September 15 to cause Dow Jones to plunge over 500 points (Singh, 2024). Major banks also turned insolvent, and the financial market had experienced one of the greatest crashes in history.
2018 - 2019	US–China Trade War due to Trump’s tariffs	The U.S. tariffs on Chinese goods in 2018 triggered a period of heightened volatility in the U.S. stock market. Investors grew concerned that escalating trade tensions would disrupt global supply chains, raise production costs, and dampen corporate earnings. Major indices such as the S&P 500 and Dow Jones Industrial Average declined sharply during tariff announcement periods, reflecting increased uncertainty and risk aversion (Chen et al., 2023).
2020 - 2021	COVID-19 Pandemic Crash	The COVID-19 shock triggered a surge in U.S. market volatility and a higher probability of adverse market states across major indices, including the S&P 500 (Basuony et al., 2021), as lockdowns constrained economic activity and heightened uncertainty. However, the technology sector flourished. The rapid shift to remote work accelerated demand for PCs and cloud infrastructure, lifting earnings expectations and valuations across tech (“Impact of Coronavirus Pandemic Crisis on Technologies and Cloud Computing Applications,” 2020). This dynamic is reflected in AMD’s post-2019 trajectory, which illustrates the sector’s outsized gains during the pandemic period.
2022 - 2023	Russia-Ukraine War	The Russia–Ukraine invasion in February 2022 triggered a risk-off shock in U.S. equities: major indices fell sharply at the open as investors priced in sanctions, surging energy costs, and weaker growth prospects. Crude oil quickly moved above \$100 per barrel, intensifying inflation and margin pressures (OECD, 2022). The result was a negative market impact with pronounced sectoral dispersion, particularly along energy and trade exposures.
2023 - 2024	Post-COVID AI-Driven Economy	The formal declaration by WHO on 5 May 2023 ending COVID-19 as a global emergency marked a broader pivot toward economic normalisation, reducing pandemic-related risk premia (Sarker et al., 2023). Meanwhile, the combination of easing monetary policy and cooling inflation expectations fueled renewed investor confidence. Capital rotated toward high-growth sectors, particularly technology and AI. This shift was powered by breakthroughs and enthusiasm around AI infrastructure, with companies like Microsoft and Nvidia becoming marquee players, driving outsized index gains that are reflected in the market index (Wearden, 2023).

Fig 1.3: Table of events

Figure 1.3 describes events that reflected the most significant price changes from December 2004 to December 2024. A more detailed explanation of the unusual price movements is provided in the Appendix.

The market index, which reflects overall financial market performance, can be seen as mainly influenced by macroeconomic conditions and global events involving major financial institutions and the U.S. Federal Reserve, which acts as the global financial powerhouse. The state of the economy therefore closely affects the movements of financial markets.

For companies, AMD, a microchip company that is highly innovative in the technology industry, is much more sensitive to global macroeconomic influences, and hence tends to experience larger fluctuations in its stock price. In contrast, the other two companies, XOM in the natural resources sector and JNJ in the pharmaceutical industry, operate in more stable sectors that are less sensitive to economic changes compared to AMD. JNJ is relatively sensitive to periods where pandemic happens such as during Covid-19, which is clearly seen in Fig 1.1 that has a plummet from 2020 till 2021 and then slowly has a recovery phase.

Stock with the Highest Returns

As shown in Figure 1.1, AMD recorded the highest returns, reflected by the largest increase between its ending (31/12/24) and initial (31/12/04) stock prices. As a strong player in the microchip industry alongside companies like Nvidia and Microsoft, AMD benefited from the strong performance of technology and internet-related stocks. Consequently, it outperformed more stable, defensive stocks such as XOM and JNJ over the period.

The Holding Period Return (HPR) is another measure used to assess which stock delivers the highest overall return. It represents the total realized return of a stock over the investment period and is calculated using the following formula:

$$HPR = \left[\frac{(\text{End of period value} - \text{Initial Value}) + \text{Income}}{\text{Initial Value}} \right] * 100$$

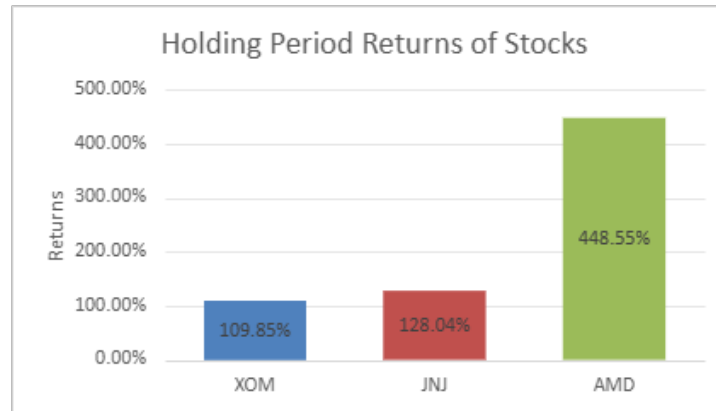


Fig 1.4: Holding period return (HPR) of the three stocks

Since dividend data is unavailable, income was assumed to be zero. Based on Figure 1.4, AMD achieved the highest HPR of 448.55% from December 2004 to December 2024. This result reinforces the earlier observation that AMD generated the greatest overall returns among the stocks analysed.

Stock with the Highest Risk

A stock's risk is broadly defined as the likelihood that its actual return deviates from the expected return. To measure this, standard deviation (SD) is commonly used as it reflects the volatility of an asset's returns relative to its historical average. A higher SD indicates greater volatility, implying a higher level of risk.

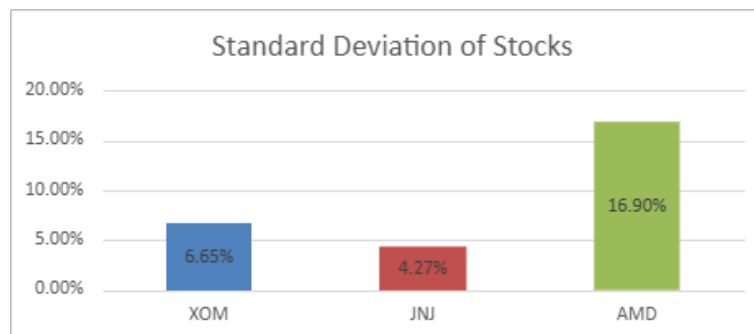


Fig 1.5: Risk levels of the three stocks

Figure 1.5 illustrates the risk level of the three stocks, calculated using the SD formula in Excel (see Task 2). AMD shows the highest level of risk, with a SD of 16.90%, a value significantly higher than its peers. Alternatively, risk can also be observed visually in Figure 1.1, where AMD's

price movements display the widest range, as marked by the sharpest spikes and dips. This suggests that AMD has the highest volatility and hence, the greatest risk.

TASK 2: Calculate mean returns and standard deviation of returns

	XOM	JNJ	AMD	Market Index	US T-Bills
Arithmetic Returns	0.5280%	0.4355%	2.1033%	0.7550%	0.1286%
Geometric Returns	0.3093%	0.3441%	0.7117%	0.6603%	0.1285%
Standard Deviation	6.6601%	4.2830%	16.9340%	4.3307%	0.1575%
Difference	0.2187%	0.0915%	1.3916%	0.0946%	0.0001%

Fig 2: Monthly geometric returns, arithmetic returns and standard deviation of the three stocks, Market Index and US T-bills

Geometric and Arithmetic Returns

Arithmetic returns represent the simple average of a stock's return over multiple periods, while geometric returns capture the time-weighted rate of return, accounting for compounding effects across periods. Since geometric returns incorporate serial correlation, also known as the effects of compounding, they provide a more accurate measure of a stock's long-term performance compared to arithmetic returns.

How Volatility affects Returns differences

The difference between geometric and arithmetic mean returns arises from fluctuations in stock prices, also known as volatility (standard deviation). The greater the volatility of a stock's returns, the larger the difference. When the arithmetic and geometric mean returns are equal, the difference is zero, indicating no volatility in the stock's performance.

As shown in Figure 2, JNJ exhibits the lowest standard deviation of 4.2830% and the smallest difference between arithmetic and geometric returns at 0.0915%. On the other hand, AMD shows the highest level of volatility among the assets, with a standard deviation of 16.9340% and the largest returns difference at 1.3916%.

This observation shows that stable stocks such as JNJ and the market index display minimal price fluctuations, resulting in nearly identical arithmetic and geometric returns. In contrast, highly volatile stocks like AMD show a much wider difference between the two measures.

This trend demonstrates that as volatility increases, compounded returns diverge more significantly from simple averages, reinforcing the importance of using geometric returns when evaluating long-term investment performance.

TASK 3: Calculate Sharpe ratios

Arithmetic Mean Risk-Free Rate

As seen in Figure 2, US T-bills' monthly arithmetic mean risk-free rate was calculated to be 0.1286%.

Sharpe ratio

The Sharpe ratio measures an investment's risk-adjusted performance by quantifying the amount of excess return earned for each unit of risk taken. It helps investors understand if a higher return is due to taking on excessive risk or making smart investment decisions.

The Sharpe ratio is derived by comparing a fund's historical or projected returns against a benchmark (risk-free rate), adjusted for the variability of those returns. It is calculated as follows:

$$\text{Sharpe ratio} = (R_p - R_f) / \sigma_p$$

Where:

R_p = return of the stock or the portfolio

R_f = Risk-free rate

σ_p = Stock or Portfolio' standard deviation

Sharpe ratio Analysis – XOM, JNJ, AMD and the Market Index

	XOM	JNJ	AMD	Market Index
Excess Returns	0.3995%	0.3071%	1.9749%	0.6265%
Sharpe Ratio	0.0600	0.0717	0.1166	0.1447

Fig 3: Sharpe ratios for the three stocks and Market Index

A higher Sharpe ratio generally indicates better risk-adjusted returns, meaning the asset delivers more excess returns for each unit of risk taken. However, a higher Sharpe ratio does not always necessarily imply the asset is less risky overall, as it may be influenced by short-term market movements or return volatility. As shown in Figure 3, the market index offers the highest risk-adjusted returns, with a Sharpe ratio of 0.1447. Therefore, if an investor could only invest in a single asset, the market index would be the most favourable choice from a risk-adjusted return perspective.

TASK 4: Calculate covariance and correlation matrices

Covariance and Correlation Analysis of Stock Returns

Covariance Matrix				Correlation Matrix			
	XOM	JNJ	AMD		XOM	JNJ	AMD
XOM	0.0044			XOM	1.0000		
JNJ	0.0010	0.0018		JNJ	0.3552	1.0000	
AMD	0.0015	0.0014	0.0287	AMD	0.1357	0.1922	1.0000

Fig 4.1 and 4.2: Covariance and correlation matrix between the three stock returns

Covariance measures the direction of the relationship between two variables, indicating if they tend to move together or in opposite directions, while correlation measures both the strength and direction of a linear relationship, providing a standardised value between -1 and +1.

The correlation matrix indicates that all pairwise correlations are relatively low, highlighting opportunities for portfolio diversification. Notably, AMD and XOM have the lowest correlation coefficient of 0.1357, suggesting that their returns move more independently than other pairs.

Risk Reduction through Diversification

The benefits of diversification increase as the correlation between assets decreases. A lower correlation means the asset's returns are less likely to move together, which helps reduce portfolio risk by offsetting individual fluctuations – thereby minimising idiosyncratic risk.

In the case that an investor only invests in AMD stock, adding the XOM stock is expected to provide greater risk reduction and enhance the benefits of diversification. This is because AMD has a lower correlation with XOM (0.1357) compared to JNJ (0.1922) as seen in Figure 4.2.

TASK 5: Calculate mean variance optimal weights

Expected Return and Variance of a Risky Portfolio

The expected return of a risky portfolio is calculated as the weighted sum of the individual returns of all the stocks within the portfolio. To compute the portfolio variance, both the individual variances of the stocks and the covariances between each pair of stocks are considered. In Excel, the following formulas are used for a 3-stock portfolio consisting of XOM, JNJ, and AMD:

$$E(r_p) = w_1 E(r_1) + w_2 E(r_2) + w_3 E(r_3)$$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2 w_1 w_2 \text{Cov}(r_1, r_2) + 2 w_1 w_3 \text{Cov}(r_1, r_3) + 2 w_2 w_3 \text{Cov}(r_2, r_3)$$

Portfolio Optimisation: With and Without Leverage

	Case 1: Short selling and leverage allowed	Case 2: Short selling and leverage NOT allowed
Weight of XOM	0.2322	0.2322
Weight of JNJ	0.4640	0.4640
Weight of AMD	0.3038	0.3038
Total Weight	1	1
Expected Portfolio Return	0.9637%	0.9637%
Variance of Portfolio	0.4108%	0.4108%
Sharpe Ratio	0.1303	0.1303

Fig 5.1: Comparison between Case 1 and 2 of optimal risky portfolios

Both Case 1 (allowing short selling and leverage) and Case 2 (long-only, no leverage) were optimised using Excel's solver to maximise the Sharpe ratio, based on the portfolio expected return and variance. In both cases, the sum of the portfolio weights was constrained to equal 1 ($w_{XOM} + w_{JNJ} + w_{AMD} = 1$). Case 2 further restricted each weight to lie between 0 and 1 (*i.e.*, $0 \leq w_{XOM} \leq 1$, $0 \leq w_{JNJ} \leq 1$, $0 \leq w_{AMD} \leq 1$). Despite these additional constraints, the optimal solution was identical in both scenarios: 23.22% in XOM, 46.40% in JNJ, and 30.38% in AMD. This outcome arose because the unconstrained optimisation naturally produced a long-only solution, rendering the additional constraints redundant. The resulting portfolio achieved an

expected monthly return of approximately 0.9637%, a standard deviation of 6.409%, and a Sharpe ratio of 0.1303, indicating that leverage is not necessary to achieve the optimal risk-return trade-off.

In theory, allowing margin trading and short selling gives investors a larger opportunity set, which can increase the Sharpe ratio. This is because the true tangency portfolio may lie outside the constrained set. Constraints like full investment and no short-selling limit the feasible portfolio space, often forcing the optimiser to settle for a suboptimal portfolio.

In this specific case, the optimal risky portfolio lies directly on the constrained efficient frontier and coincides with the tangency point of the capital market line. This means the investor is already holding the unique portfolio that delivers the highest possible Sharpe ratio without needing leverage. As a result, allowing margin trading and short selling did not lead to an improvement.

Impact of Diversification on Sharpe ratio

	AMD	Optimal Risky Portfolio (Case 2)
Sharpe Ratio	0.1166	0.1303

Fig 5.2: Comparison of Sharpe ratio between AMD and Optimal Risky Portfolio (Case 2)

The Sharpe ratio for AMD alone is 0.1166, whereas the optimal risky portfolio, comprising only long, unleveraged positions, has a Sharpe ratio of 0.1303 – an improvement of 0.0137. While the numerical difference appears small, it represents a significant 12% improvement in risk-adjusted returns. This gain highlights the power of diversification: because XOM, JNJ, and AMD have low correlations (well below 1), combining them reduces portfolio volatility more than it reduces expected return. Even a slight reduction in risk can result in a meaningfully better risk-adjusted outcome compared to holding a single stock on its own. This leads to a more efficient portfolio that shifts upward along the efficient frontier, offering a higher return per unit of risk.

TASK 6: Calculate the Capital Allocation Line (CAL)

A complete portfolio combines a risk-free asset with a single risky portfolio. To match AMD's risk level, the Capital Allocation Line (CAL) is used to illustrate all possible combinations and their corresponding Sharpe ratios. In this analysis, two scenarios are compared: the optimal risky portfolio (XOM, JNJ, and AMD) and the market index, each adjusted to achieve the same level of risk as AMD.

How Capital is Allocated at AMD's Risk

	Weights	Per \$10,000 invested
XOM	0.6133	\$ 6,133.40
JNJ	1.2259	\$ 12,258.96
AMD	0.8027	\$ 8,026.85
Risk-free	-1.6419	-\$ 16,419.21

Fig 6.1: Case 1 – Final allocations in the complete portfolio

	Weights	Per \$10,000 invested
Market Index	3.9102	\$ 39,102.11
Risk-Free	-2.9102	-\$ 29,102.11

Fig 6.2: Case 2 – Final allocations in the complete portfolio

The cases shown in Figures 6.1 and 6.2 are constructed to match AMD's risk level.

In Case 1 (optimal risky portfolio), the CAL suggests allocating 264.19% (shown by the weights) to the portfolio and borrowing 164.19% at the risk-free rate. With the initial investment of \$10,000, this translates to \$26,419 invested in the optimal risky portfolio (\$6,100 in XOM, \$12,200 in JNJ, \$8,000 in AMD) and -\$16,419 in T-bills.

In Case 2 (market index), the CAL suggests allocating 391.02% to the index and borrowing 291.02% at the risk-free rate. This equates to \$39,102 invested in the market index and -\$29,902 in T-bills.

These leveraged allocations allow both portfolios to replicate AMD's risk level efficiently by using borrowing at the risk-free rate to scale returns along the CAL.

Expected Wealth in 20 years: Optimal risky portfolio vs AMD

	Expected Value
Complete Portfolio	\$ 2,545,432.11
AMD	\$ 1,477,693.85
Difference	\$ 1,067,738.26

Fig 6.3: Expected value of portfolio in 20 years, given initial investment of \$10,000 (Optimal Risky Portfolio)

Figure 6.3 illustrates how the optimal risky portfolio combined with T-bills is projected to grow to \$2.55m over 20 years, compared to \$1.48m for AMD alone – a gain of \$1.07m. Both portfolios are set to AMD’s volatility, meaning the performance difference stems from a higher return per unit of risk.

The optimal portfolio has a higher Sharpe ratio because diversification across XOM, JNJ, and AMD reduces stock-specific noise while preserving overall return potential. On the CAL this is represented by a steeper slope; when scaled to AMD’s risk it delivers a higher monthly excess return, which compounds to the \$1.07m advantage.

Expected Wealth in 20 years: Market index vs AMD

	Expected Value
Complete Portfolio	\$ 4,495,817.08
AMD	\$ 1,477,693.85
Difference	\$ 3,018,123.24

Fig 6.4: Expected value of portfolio in 20 years, given initial investment of \$10,000 (Market Index)

Figure 6.4 illustrates that when the market index serves as the risky portfolio, the complete portfolio is projected to grow to \$4.47m over 20 years, whilst matching AMD’s volatility. This is \$3.02m more than AMD alone, due to the market index’s higher risk-adjusted return, reflected in its steeper CAL slope.

The main difference is that the market index offers even stronger compensation for risk, which is reflected in its steeper CAL slope, so the same target level of volatility produces a higher average monthly return and greater wealth accumulation over time.

Why the Market Index Outperforms the Optimal Risky Portfolio

The difference in additional wealth between the optimal risky portfolio and the market index can be explained by the slope of the CAL in each case. The market index's slope sits higher, which means a larger increase in expected returns per unit of risk at the AMD volatility level. This advantage likely results from broader diversification, less individual stock risk, and more stable long-term performance.

TASK 7: Calculate CAPM alpha and beta

Fixed weights are used for the optimal risky portfolio to generate a single return series for CAPM testing. This approach establishes a clear policy portfolio, isolates average performance, and minimises noise from frequent rebalancing or transaction costs. It also aligns with how passive and benchmark portfolios are typically evaluated.

That said, in practice, fixed weights are a simplification. Risks and correlations change over time, so portfolio managers typically rebalance as conditions shift. A constant mix can become sub-optimal or riskier in new environments. While fixed weights are valuable for benchmarking purposes, they do not fully capture the dynamic management of active portfolios in real-world settings.

	Optimal	XOM	JNJ	AMD
Alpha	0.0015	-0.0009	-0.0004	0.0061
t-stat (Alpha)	0.5217	-0.2278	-0.1900	0.6670
Beta	1.0998	0.7744	0.5601	2.1727
t-stat (Beta)	17.1303	8.9923	10.6000	10.3104
R²	0.5522	0.2536	0.3207	0.3088

Fig 7: Alpha/Beta Regression Summary

Figure 7 shows that AMD has the highest beta among the assets, amplifying market movement, while JNJ stands out for its low beta and more defensive profile. The optimal risky portfolio lies slightly above market sensitivity, with a beta nearly equal to the market index, reflecting a well-diversified mix designed for risk efficiency. The portfolio's R^2 value of 0.5522 indicates a considerable portion of the variation in monthly returns, but individual risks and unique asset attributes continue to matter, especially for single stocks.

Assessing the Attractiveness of Switching to the Optimal Risky Portfolio

Switching looks unattractive. The optimal risky portfolio's alpha is 0.15% per month with a t-stat of 0.52, so it is statistically indistinguishable from zero; its beta is 1.10 (slightly more market risk) and R^2 is 0.55. In short, there's no reliable risk-adjusted outperformance versus the index. The market index remains the default unless an investor specifically wants the XOM/JNJ/AMD tilts and accepts tracking error.

Interpreting Results in the Context of Weak-Form Market Efficiency

The results are consistent with weak-form efficiency: past prices do not yield reliable, risk-adjusted profits. The small alpha values observed for XOM and JNJ suggest that market prices already incorporate all relevant information from past returns, making it difficult for investors to gain an advantage by relying on historical price patterns. The positive alphas calculated for the optimal risky portfolio and AMD likely reflect the specific sample period or risks not fully captured in the analysis, rather than a genuine opportunity to exploit past price information. Overall, these findings reinforce the idea that broad market trends and systematic risk remain the primary drivers of returns, and that opportunities for consistent excess gains are limited in an efficient market.

Appendix

The table below comprises of a summary of all events that led to unusual price changes (Task 1).

Year	Event	Description
2007-2009	2008 Global Financial Crisis	The 2008 American mortgage crisis triggered a worldwide financial collapse, with Lehman Brother's bankruptcy on September 15 to cause Dow Jones to plunge over 500 points (Singh, 2024). Major banks also turned insolvent, and the financial market had experienced one of the greatest crashes in history.
2010-2011	European Sovereign Debt Crisis	Recovery momentum was challenged when Greece's near default in 2010 caused broad European debt crisis which involves many countries like Portugal, Ireland, Italy and Spain. This causes intensified fear among the market after S&P downgraded U.S. credit rating from AAA to AA+ on August 8, 2011, which caused 6-7% plunge in U.S. indices in a single day (Wearden, 2011).
2015	China Stock Market Crash	The nearly 9% collapse of Chinese stocks in 2015 triggered a widespread investor fear, which rippled across global markets, including the US market. Overall, the US stock market fell, with Dow Jones plunging ~3.55% while the S&P 500 closing down ~3.93% (News, 2015).
2018-2019	US-China Trade War due to Trump's tariffs	The U.S. tariffs on Chinese goods in 2018 triggered a period of heightened volatility in the U.S. stock market. Investors grew concerned that escalating trade tensions would disrupt global supply chains, raise production costs, and dampen corporate earnings. Major indices such as the S&P 500 and Dow Jones Industrial Average declined sharply during tariff announcement periods, reflecting increased uncertainty and risk aversion (Chen et al., 2023).
2020-2021	COVID-19 Pandemic Crash	The COVID-19 shock triggered a surge in U.S. market volatility and a higher probability of adverse market states across major indices, including the S&P 500 (Basuony et al., 2021), as lockdowns constrained economic activity and heightened uncertainty. However, the technology sector flourished. The rapid shift to remote work accelerated demand for PCs and cloud infrastructure, lifting earnings expectations and valuations across tech ("Impact of Coronavirus Pandemic Crisis on Technologies and Cloud Computing Applications," 2020). This dynamic is reflected in AMD's post-2019 trajectory, which illustrates the sector's outsized gains during the pandemic period.
2022-2023	Russia-Ukraine War	The Russia-Ukraine invasion in February 2022 triggered a risk-off shock in U.S. equities: major indices fell sharply at the open as investors priced in sanctions, surging energy costs, and weaker growth prospects. Crude oil quickly moved above \$100

		per barrel, intensifying inflation and margin pressures (OECD, 2022). The result was a negative market impact with pronounced sectoral dispersion, particularly along energy and trade exposures.
2023 onwards	Post-COVID AI-Driven Economy	The formal declaration by WHO on 5 May 2023 ending COVID-19 as a global emergency marked a broader pivot toward economic normalisation, reducing pandemic-related risk premia (Sarker et al., 2023). Meanwhile, the combination of easing monetary policy and cooling inflation expectations fueled renewed investor confidence. Capital rotated toward high-growth sectors, particularly technology and AI. This shift was powered by breakthroughs and enthusiasm around AI infrastructure, with companies like Microsoft and Nvidia becoming marquee players, driving outsized index gains that are reflected in the market index (Wearden, 2023).

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